

RAREȘ ANDREI STANA

STUDENT / SOFTWARE DEVELOPER

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PROFESSIONAL EXPERIENCE

Software Developer, Soft31

July 2024 – October 2025

Exammy.ai - AI-Powered Test Generator for Medical Schools

- Led the development of an automated test question generation platform designed for medical institutions.
- Used Meta Llama 3 with retrieval-augmented generation (RAG) over curated medical syllabi/guidelines to produce high-quality, clinically relevant exam content, significantly reducing manual workload for educators.
- Built the platform using Python and Streamlit, ensuring intuitive UI and operational efficiency.
- Helped streamline assessments while maintaining academic rigor and offering flexible use for medical faculty.

SNIAS Virtual Assistant - Romanian Ministry of Labor (ANPIS)

- Sole developer responsible for the full-stack implementation of a chatbot that assists Romanian citizens in navigating social welfare services.
- Developed and integrated the AI chatbot using Typebot + Angular (HTML/CSS/JS), powered by OpenAI (GPT) for NLU/response generation and orchestrated via n8n workflows.
- Created a user-friendly flow based on complex institutional policy logic, covering benefits such as child support, disability aid, and heating subsidies.
- Enabled features like human redirection, benefit application guidance, and email transcript delivery.

PROJECTS

Ontology-Based Conversational Hotel Recommendation System

Vrije Universiteit Amsterdam

- Designed and implemented a conversational recommendation system combining ontology-based reasoning with Large Language Models to answer accommodation-related queries in a factual and explainable way.
- Implemented an ontology-driven query answering pipeline, where an LLM was strictly used for natural language query extraction, while all factual inference was performed by an OWL reasoner to prevent hallucinations.
- Translated user queries into Manchester OWL class expressions and evaluated them using a reasoner to infer valid entities based on multiple constraints (location, type, proximity).
- Evaluated the system using simple and complex dialogue scenarios, demonstrating reliable reasoning, correct fallback behavior when data was missing, and high factual consistency.

SplitSmart - AI-Powered Receipt Splitting App

Vrije Universiteit Amsterdam / bunq + AWS Hackathon

- Built an AI-powered receipt splitting app for flatmates, allowing users to upload receipts, extract line items, and automatically calculate who owes what.
- Developed a React + Vite frontend and FastAPI backend, with receipt data storage, flatmate management, spending history, and shared/personal item assignment.
- Integrated Claude via AWS Bedrock to extract and categorize receipt items, with manual review and correction support to improve splitting accuracy.
- Added bunq sandbox payment request flow concepts, store detection, and price comparison features to make shared grocery expenses easier to manage.

Predator-Prey Agent-Based Simulation

Vrije Universiteit Amsterdam

- Built an agent-based model in Python (Violet Simulator) to study ecosystem resilience under environmental crises.
- Simulated predator-prey dynamics (rabbits/foxes/grass) with energy, reproduction, and mortality rules.
- Analyzed population outcomes across 100+ runs, showing how short-term disasters reshape ecological balance.

SKILLS

Languages: Python, JavaScript, SQL

Frameworks: FastAPI, Flask, React, Angular, Streamlit

AI/ML: LLMs, RAG, NLP, TensorFlow/Keras, LangChain, Claude, OpenAI API, AWS Bedrock

Data: PostgreSQL, Elasticsearch, JSON, REST APIs

Tools & Platforms: Git, Docker, AWS, n8n, Typebot, Kibana, Protégé, OWL Reasoner, Vite

EDUCATION

Bachelor in Artificial Intelligence

September 2023 – June 2026

Vrije Universiteit Amsterdam

- Relevant coursework: Machine Learning, Data Structures & Algorithms, Probability & Statistics, Intelligent Systems

ADDITIONAL INFORMATION

- **Languages:** English, Romanian.
- **Certificates:** Oracle Certified Associate (OCA), European Computer Driving Licence (ECDL), Ethereum and Solidity: The Complete Developer's Guide (Udemy, 2023).